FloodGuard AI Backend

Technical Features and Implementation Overview

Backend	FastAPI / Python
Document version	1.3
Backend version	0.1.0
Last updated	1 October 2026
Default data mode	`live`
Primary risk model	`rule_based_v1` on live inputs

> **Model status:** Two real-data XGBoost research models have been trained and geographically backtested. Both failed the documented promotion gate and are not deployed. Production remains on the declared rule engine with live inputs.

1. Purpose and Scope

FloodGuard AI is a backend for flood-condition monitoring, national district screening, risk inference, prediction history, operational state, incidents, citizen reports, emergency resources, and downstream automation context. It exposes

versioned REST APIs under `/api/v1`.

> **Live national weather batches + station river discharge → district screening and detailed XGBoost risk → cached GIS layer + persisted predictions and events**

2. Runtime Architecture

Layer	Implementation	Responsibility
API	FastAPI	Routing, validation, OpenAPI documentation, CORS, and structured errors.
Providers	Async `httpx` adapters	Retrieve and normalize weather and river data behind common interfaces.
Risk	XGBoost with configurable rule engine	Calculate flood probability, severity, status, factors, and confidence. Production defaults do not silently change models.
Services	Python service classes	Coordinate providers, inference, persistence, trends, events, and monitoring state.
National GIS	GeoJSON + cached batch processing	Serve Pakistan district geometry, weather screening, viewport filtering, provenance, and freshness metadata.
Persistence	SQLAlchemy	Predictions, incidents, citizen reports, system events, and system state.
Database	SQLite by default	Local persistence; configurable for another SQLAlchemy database URL.

3. Data Sources and Modes

Mode	Weather	River	Intended use
`live`	Open-Meteo Forecast API	Open-Meteo Flood API / GloFAS	Default operational mode.
`hybrid`	Same live provider	Same live provider	Backward-compatible alias.
`simulation`	Deterministic scenarios	Deterministic scenarios	Development and automated tests only; rejected unless `ENABLE_SIMULATION=true`.

Weather normalization

The live provider requests hourly precipitation, temperature, relative humidity, and 0-7 cm soil moisture. It calculates current, previous 1/6/24-hour, and forecast 6/24-hour rainfall totals.

River normalization

The live river provider uses GloFAS daily modeled discharge at configured river coordinates. It returns current discharge, daily change, next-day forecast, and a simple rising/normal status.

> **River-level constraint:** GloFAS provides modeled discharge, not measured gauge height. Live responses therefore return level-related fields as `null`. FloodGuard does not fabricate a water level.

4. Flood-Risk Model

Model	Purpose and status
`rule_based_v1`	Current production model. Applies documented hydrometeorological weights to live weather and modeled discharge. It is deterministic and not a learned model.
`xgboost_v1`	Development-only regressor trained on 6,000 synthetic rows. Disabled in production and never represented as historical training.
`xgboost_pakistan_gfd_v1`	Real-data binary classifier trained on historical Pakistan event windows. Failed event and geographic validation; retained as a rejected research artifact and not used by the API.
`xgboost_pakistan_desinventar_v2`	Real-data district-date classifier with richer antecedent rainfall and normalized discharge features. Failed district, geographic, and temporal validation; retained as a rejected research artifact and not used by the API.
`xgboost_lower_indus_pilot_v1`	Explicit partial-coverage experiment for 18 districts in six contiguous Lower Indus geographic blocks. Failed district, block, and temporal validation; not used by the API.

Synthetic development dataset

The generator in `app/ml/synthetic_data.py` remains only for development and regression tests:

- Rainfall, river conditions, discharge, discharge change, and soil moisture are sampled from bounded synthetic ranges.
- Discharge is correlated with synthetic river level; discharge change is correlated with synthetic rise rate.
- The probability label is a weighted hydrometeorological score plus a high-rain/high-river interaction and small random noise.
- 35% of rows have level-related inputs removed to train the model to operate when only discharge is available.

This synthetic artifact is disabled by default with `ML_MODEL_ENABLED=false`.

Real historical dataset

Item	Implementation
Flood-event source	Global Flood Database v1 event catalogue produced from MODIS flood mapping and Dartmouth Flood Observatory events.
Coverage	28 events associated with Pakistan between 2003 and 2018; 2,410 daily rows, including 730 event-window rows.
Weather source	Open-Meteo Historical Weather API: hourly precipitation and 0-7 cm soil moisture, aggregated into daily model features.
Hydrology source	Open-Meteo Flood API / GloFAS historical modeled daily discharge and day-to-day discharge change.
Label	1 when the date is inside the DFO/GFD event interval at the event centroid; 0 for matched 31-day pre/post-event windows.
Missing fields	Measured gauge height, warning-level ratio, and gauge rise rate remain missing because no authorized historical gauge series is connected.
Dataset artifact	`app/ml/data/pakistan_real_flood_training.csv`.
Rebuild command	`python -m app.ml.train_real_model --rebuild`.

> **Label limitation:** the event catalogue confirms mapped inundation somewhere in the event footprint. It does not prove that the event centroid flooded on every date in the full event interval. The next-day observed rainfall is used as a historical proxy for the operational 24-hour forecast feature. These limitations make this useful for research, not operational certification.

Model input features

Feature	Operational source	Notes
`rainfall_last_1h`	Open-Meteo weather	Previous-hour precipitation total.
`rainfall_last_6h`	Open-Meteo weather	Six-hour accumulated precipitation.
`rainfall_last_24h`	Open-Meteo weather	Twenty-four-hour accumulated precipitation.
`forecast_rainfall_24h`	Open-Meteo weather	Next 24-hour precipitation forecast.
`river_level`	Authorized gauge feed when available	Missing in current live mode and passed as an XGBoost missing value.
`river_level_relative_to_warning`	Derived from gauge values	Missing when observed level and warning level are unavailable.
`river_rise_rate`	Authorized gauge feed when available	Missing in current live mode.
`river_discharge`	GloFAS	Modeled daily discharge in cubic metres per second.
`river_discharge_change`	Derived from GloFAS	Current discharge minus previous-day discharge.
`soil_moisture`	Open-Meteo weather	0-7 cm volumetric soil moisture converted to percent.

Severity classification

Probability	Severity	Status
< 0.38	LOW	NORMAL
0.38 to < 0.58	MEDIUM	UNDER_VERIFICATION
0.58 to < 0.78	HIGH	PREDICTED
≥ 0.78	CRITICAL	PREDICTED

Failure policy

Production uses `ALLOW_RULE_BASED_FALLBACK=false`. If the enabled ML artifact is missing, corrupt, or fails inference, the API returns structured HTTP 503 instead of silently changing models. A rule-based fallback can be enabled deliberately for a separate degraded-mode deployment. Provider failures never substitute simulation data and return structured HTTP 502.

Production also defaults to `ML_MODEL_ENABLED=false` because the bundled XGBoost artifact is trained on synthetic development data. In this mode `rule_based_v1` is the declared primary model operating on live inputs, not a silent fallback. Enable the bundled artifact only for development, or replace it with a historically trained and validated artifact before operational ML use.

Real-data research run

The classifier uses `binary:logistic`, depth 3, learning rate 0.04, row subsampling 0.85, column subsampling 0.90, early stopping, and seed 42. Two independent holdouts are required:

Metric	Event holdout	Geographic holdout
Test rows	583	682
ROC-AUC	0.5936	0.3836
Accuracy at 0.5	0.6175	0.6891
Balanced accuracy at 0.5	0.5000	0.5197
Precision at 0.5	0.0000	0.4038
Recall at 0.5	0.0000	0.1040
F1 at 0.5	0.0000	0.1654
Brier score	0.2446	0.2219

Event holdout keeps complete events in one split only. Geographic holdout assigns all rows in each 2-degree latitude/longitude block to one split only; 15 geographic blocks are represented. This tests transfer to unseen areas.

Promotion requires both holdouts to achieve ROC-AUC at least 0.75, precision at least 0.60, and recall at least 0.60. The model failed. Ordinary accuracy is not accepted by itself because majority non-flood predictions can inflate it. Status is `rejected_holdout_quality`, `deployment_eligible=false`, and production remains on `rule_based_v1`.

Full provenance, exact split membership, metrics, limitations, and acceptance gates are stored in `app/ml/artifacts/flood_risk_xgboost_real.model-card.json`.

DesInventar district-date research run

The second experiment uses the official UNDRR DesInventar Pakistan export. The cleaning pipeline keeps `FLOOD` and `FLASH FLOOD`, validates dates from 2000 onward, applies explicit historical district aliases, rejects unknown districts, and deduplicates district-date labels. It produced 1,704 cleaned positive district-dates from 2,912 raw flood records. Year 2007 was excluded completely from model rows because the free historical provider quota ended before national covariate coverage was complete; excluding the whole year avoids partial-district selection bias.

Item	Value
Training rows	10,097: 1,584 positive and 8,513 matched controls.
Spatial coverage	67 mapped districts in 23 two-degree geographic blocks.
Controls	Same-district dates at +/-7, 14, 21, and 28 days, excluding +/-3 days around a recorded event.
Weather	Open-Meteo historical daily precipitation and 0-7 cm soil moisture.
Hydrology	Open-Meteo Flood API / GloFAS modeled daily discharge.
Artifacts	`app/ml/data/pakistan_desinventar_training.csv`, quality report, model JSON, and model card.
Rebuild	`python -m app.ml.train_desinventar_model --rebuild`.

XGBoost uses binary logistic loss, class weighting, 500 maximum rounds, early stopping, row/column subsampling, seed 42, and a validation-only grid over depth 2/3/4, learning rate 0.03/0.06, and minimum child weight 5/15. The classification threshold is selected only from validation predictions.

Metric	District holdout	Geographic holdout	Temporal holdout
Test rows / positives	1,224 / 153	2,677 / 584	679 / 100
ROC-AUC	0.6815	0.5704	0.7029
Selected threshold	0.42	0.50	0.62

Metric	District holdout	Geographic holdout	Temporal holdout
Precision	0.1765	0.2654	0.3396
Recall	0.8039	0.2877	0.3600
F1	0.2894	0.2761	0.3495
Balanced accuracy	0.6340	0.5328	0.6196
Brier score	0.2019	0.2129	0.2350

The richer feature set improved spatial ROC-AUC over the first experiment from 0.3836 to 0.5704, but it still fails the gate of ROC-AUC 0.75, precision 0.60, and recall 0.60 on every holdout. Status is `rejected_holdout_quality`; it is not loaded by the API.

V2 feature availability and sufficiency

Feature group	Historical training	Current live availability
Rainfall 1/3/7/14/30 days	Available from Open-Meteo reanalysis.	Only the latest 24 hours are currently fetched. Longer windows require persisted observations or a larger live request.
Next-day rainfall	Next observed day is a forecast proxy.	Available as an actual Open-Meteo forecast.
0-7 cm soil moisture	Available from Open-Meteo reanalysis.	Available from Open-Meteo forecast/current data.
Discharge, daily change	Available from GloFAS reanalysis.	Available as modeled discharge, not a gauge observation.
7/30-day discharge ratios and 30-day z-score	Available from GloFAS history.	The live provider fetches seven past days; 30-day features require an extended request or persisted history.
Elevation and month seasonality	Available.	Available or directly derived.
Gauge height, warning threshold, rise rate	Unavailable in the connected public datasets.	Unavailable until an authorized FFD/WAPDA gauge feed and station thresholds are integrated.

> **Conclusion:** XGBoost capacity and tuning are not the main blocker. Geographic transfer remains weak because district representative points do not describe full catchments, DesInventar is an impact-reporting catalogue rather than continuous inundation truth, and measured gauge levels, station thresholds, catchment rainfall, terrain slope, river distance, drainage capacity, and flood-protection context are absent. Recommended next data work is authority gauge integration, catchment/raster aggregation, static susceptibility layers, and event-based geographic backtesting before another promotion decision.

Declared partial-coverage experiment

A smaller model was tested rather than assuming national coverage. The Lower Indus pilot was fixed before evaluation as six contiguous blocks covering Badin, Barkhan, Dadu, Ghotki, Hyderabad, Jacobabad, Jafarabad, Jhal Magsi, Kachhi, Kalat, Khuzdar, Kohlu, Lasbela, Mastung, Nasirabad, Sibi, Thatta, and Umerkot. The dataset contains 4,597 rows and 652 positive district-dates.

Validation	ROC-AUC	Precision	Recall	Decision
District holdout	0.5295	0.1479	0.6488	Failed
Leave-one-block-out	0.5567	0.1646	0.4831	Failed

Validation	ROC-AUC	Precision	Recall	Decision
Temporal holdout	0.4962	0.0000	0.0000	Failed

Each of the six blocks is used as an unseen test block once, with a different validation block and the remaining blocks used for training. The experiment shows that geographic size is not the main issue: regional labels still describe district-wide reported impacts while the predictors describe one representative point. The artifact is `rejected_holdout_quality` and cannot return predictions outside its declared coverage.

5. Trends and Adaptive Monitoring

Each prediction is compared with the latest previous prediction for the same location.

Probability change	Trend
≤ -0.05	FALLING
> -0.05 and < 0.05	STABLE
≥ 0.05 and < 0.15	RISING
≥ 0.15	RAPIDLY_RISING

Base severity	Base interval	Trend adjustment
LOW	60 minutes	RISING shortens by one tier. RAPIDLY_RISING sets 5 minutes. Downgrades respect the configured 30-minute cooldown.
MEDIUM	30 minutes	
HIGH	15 minutes	
CRITICAL	5 minutes	

6. National GIS Screening

The endpoint `GET /api/v1/gis/national-risk` returns a GeoJSON FeatureCollection covering 126 Pakistan ADM2 district geometries. The packaged simplified boundaries are from geoBoundaries PAK ADM2 (2019), distributed as public-domain data. The optional `bbox=west,south,east,north` parameter filters output to a map viewport.

Item	Implementation
Live ingestion	District representative coordinates are sent to Open-Meteo in configurable batches of 40.
Cache	Successful national layers are cached for 1,800 seconds by default. A stale successful layer is returned with a warning if a later provider refresh fails.
Screening inputs	Recent 6/24-hour rainfall, forecast 24-hour rainfall, and 0-7 cm soil moisture.
Score	`0.34*min(rain24/120,1) + 0.38*min(forecast24/100,1) + 0.18*min(soil/90,1) + 0.10*min(rain6/50,1)`, constrained to [0.01, 0.98].
Output labels	`district_weather_screening`, `district_weather_screening_v1`, and `screening_index`.
Confidence	0.62 for live screening. This is a prioritization layer, not a calibrated flood forecast.
Simulation	Scenario-dependent deterministic district scores, clearly marked with `data_mode=simulation`.

> **Interpretation:** The national district layer is weather-derived screening for prioritization. It does not infer gauge height, flood depth, inundation extent, or district exposure. The existing station endpoint remains the river-aware XGBoost prediction path.

7. Backend Feature Inventory

Feature	Implementation summary
Health and system status	Reports process status, version, environment, data mode, active scenario, and monitoring state.
Weather	Normalized current and accumulated live weather by configured location.
River conditions	Live modeled discharge, daily change, next-day forecast, and status.
Flood risk	ML probability with severity, status, trend, factors, model identity, monitoring interval, and source metadata.
National GIS risk	Cached district GeoJSON with live weather screening, viewport filtering, source timestamps, model labels, and boundary provenance.
Prediction history	Persists and filters predictions by location and limit.
Incidents	Create, retrieve, list, and update flood incidents.
Citizen reports	Create, retrieve, and list geolocated reports with pending verification state.
System events	Records risk updates, monitoring changes, incident/report activity, simulation transitions, and critical alert requirements.
Simulation	Deterministic normal-to-confirmed scenarios isolated from live operational inference.
Automation context	Provides risk, severity, status, interval, and verification/alert flags for downstream workflows.
Dashboard summary	Aggregates incident, report, prediction, and monitoring statistics.
Emergency resources	Returns approved configured shelters, hospitals, and emergency contacts. Lists remain empty until an authority dataset is connected; no placeholder facilities are shipped.
API contract	Pydantic validation, structured 404/422/502 responses, Swagger UI, ReDoc, and OpenAPI schema.

8. Stored Data

Entity	Stored information
Prediction	Probability, severity, status, confidence, trend, change, factors, model identity, source identity, interval, and timestamp.
Incident	Location, flood type, severity, lifecycle status, confidence, description, source, and timestamps.
Citizen report	Location, coordinates, description, optional image URL/depth, source, verification state, and timestamp.
System event	Event type, message, location, structured metadata, and timestamp.
System state	Active scenario/location, monitoring mode/interval, simulation time, and last monitoring change.

9. Known Limitations and Next Production Steps

1. Obtain quality-controlled gauge observations, official station warning thresholds, and verified flood extents/outcomes from FFD, WAPDA, NDMA, or provincial authorities.

2. Add catchment-average rainfall, terrain slope/elevation distribution, river distance, drainage, land cover, and flood-protection features.
3. Integrate authorized FFD/WAPDA gauge observations and station-specific flood thresholds.
4. Use station-specific rating curves only when validated; never convert discharge to level with a generic formula.
5. Replace the 2019 public-domain demonstration boundaries with NDMA-authorized current administrative boundaries.
6. Add provider retry/backoff, detailed-location caching, dependency-aware health checks, and production database migrations.
7. Move high-volume historical GIS observations to PostGIS and vector/raster tiles as data density grows.
8. Verify and routinely update static emergency-resource records.

10. Maintenance

This HTML file is the editable source for the PDF. Update it whenever providers, model features, thresholds, endpoints, persistence fields, or validation status change. Regenerate on Windows with:

```
powershell -ExecutionPolicy Bypass -File scripts\generate_backend_pdf.ps1
```

Source of truth: current FloodGuard AI backend implementation. This document describes implemented behavior; it does not certify hydrological or emergency-response accuracy.